INDIVIDUAL TECHNICAL & EMPIRICAL METHODOLOGY REPORT

Automated SEC Form 8-K Product Release Extraction Pipeline

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1. Executive Summary & Research Objective

This investigation designs, deploys, and evaluates an automated natural language processing (NLP) and information extraction pipeline targeting new-product announcements within regulatory corporate disclosures. The system processes Form 8-K filings submitted to the U.S. Securities and Exchange Commission (SEC) across the complete 2025 calendar year (January 1 to December 31, 2025). Form 8-K serves as the statutory instrument for notifying markets of unscheduled material corporate developments. While primary items govern executive turnover (Item 5.02) and earnings releases (Item 2.02), commercial technological announcements are frequently introduced under Item 7.01 (Regulation FD Disclosure) or Item 8.01 (Other Events), routinely substantiated by press releases attached as Exhibit 99.1.

The empirical corpus comprises a frozen sample of exactly 100 unique 8-K filings drawn from 20 distinct publicly traded technology and industrial entities. Using a locally hosted, quantized instruction-tuned large language model (LLM) operating inside an isolated Google Colab GPU environment under zero temperature and JSON schema constraints, the pipeline extracts entity names, ticker symbols, acceptance timestamps, verbatim product names, and concise descriptions strictly constrained under 180 characters into structured CSV and pipe-delimited datasets.

2. SEC Ingestion Architecture & Fair Access Protocols

The ingestion engine interfaces directly with official SEC EDGAR endpoints while enforcing compliance with federal automated query guidelines:

- **Entity Directory Mapping:** Central Index Keys (CIKs) are programmatically mapped from corporate equity tickers using the official SEC `company_tickers.json` directory, ensuring standard 10-digit zero-padded CIK resolution [1].
- **Discovery & Stratified Round-Robin Allocation:** Filings are gathered via the instructor-specified `browse-edgar` Atom XML feed and supplemented with official SEC Submissions API records (`data.sec.gov/submissions/CIK{cik}.json`) to ensure complete annual filing history and exact form matching. To prevent corporate reporting bias and sample distortion, a deterministic round-robin allocation draws an upper ceiling of 10 filings per entity across 20 candidate tickers (`AMSC`, `AZTA`, `CRNC`, `DAKT`, `DIOD`, `FEIM`, `IPGP`, `IRDM`, `KN`, `KOPN`, `MLAB`, `MTSI`, `NVTS`, `POWI`, `SMTC`, `SPSC`, `VREX`, `ALNT`, `CLFD`, `DGII`) before product extraction commences. This guarantees balanced coverage across mid-cap hardware, networking, semiconductor, and communication manufacturers.
- **Pacing & Protocol Compliance:** All HTTP requests declare a custom academic user-agent header (`Crispen Chachengwa academic 8-K research cris2214zw@gmail.com`), enforce client-side request pacing (minimum 350 ms spacing to adhere to the 10 request-per-second federal ceiling), leverage local SHA-256 binary caching, and implement bounded exponential backoff on transient HTTP 429 and 5xx status codes.

3. Document Preprocessing, Scope Bounding & Context Batching

Corporate 8-K submissions routinely contain ancillary exhibits irrelevant to product operations. The ingestion scope is strictly bounded to the primary Form 8-K report and its first press release attachment, designated as Exhibit 99.1 (`EX-99.1` or `EX-99.01`). Supplementary filings (e.g., credit agreements, executive compensation schedules, inline XBRL taxonomies) are explicitly cataloged as excluded exhibits within checkpoint manifests.

Document texts are parsed using BeautifulSoup (for HTML/XML markup decomposition) and `pypdf` (for unstructured PDF exhibits) [2]. Raw character streams are divided into source-linked passages of approximately 900 characters snapped cleanly at whitespace boundaries. To optimize LLM inference while avoiding silent truncation:

- **Regex Signal Triage:** Passages are scanned for corporate release signals (`\b(?:launch\w*|introduc\w*|unveil\w*|debut\w*|announc\w*|new\s+(?:product|device|version|platform)|general\s+availability|initial\s+shipments?|now\s+(?:available|shipping)|roll[ -]?out)\b`). When hits occur, the pipeline selects the matched unit along with its adjacent neighboring units (±1 unit context window).
- **Full-Text Fallback:** If no regex signal triggers across an exhibit, the pipeline does not record an automated negative; rather, it automatically reverts to full-text ingestion to guarantee recall across subtle or non-standard announcement phrasings.
- **Batch Assembly & Guardrails:** Selected units are grouped into contextual batches of ~9,000 characters with a 1-unit boundary overlap, bounded by an expanded 12-batch ceiling per filing. Excessively long or illegible image-only files trigger auditable error flags rather than silent loss.

4. LLM Named Entity Recognition & Constrained Extraction

Inference executes entirely in-runtime on an isolated Google Colab NVIDIA GPU environment using Ollama serving `qwen3:4b-instruct-2507-q4_K_M` (allocating 3.61 GiB VRAM), eliminating external proprietary API dependencies and data leakage [3], [4]. The model performs joint Named Entity Recognition (NER)—extracting `ORGANIZATION` and `PRODUCT` entity types—alongside commercial release event extraction.

To prevent hallucinations and guarantee parsing stability, Ollama structured decoding enforces an immutable JSON Schema requiring: (1) `organization` name, (2) exact contiguous `product_name`, (3) `product_description` between 1 and 179 characters, (4) valid source `evidence_ids` enum references, and (5) a boolean `needs_review` ambiguity flag. Inference runs deterministically at `temperature: 0`, `seed: 42`, an 8,192-token context window, and a 1,500-token completion ceiling.

5. Deterministic Mechanical Grounding & Decision Logic

Candidate extractions emitted by the model undergo multi-stage deterministic Python validation prior to export:

1. **Textual Verbatim Check:** The candidate `product_name` must appear verbatim within the cited passage text after unicode NFKC normalization. Invented names are immediately discarded.
2. **Entity Disambiguation:** Normalized candidate product names cannot match the reporting company's own legal name, preventing company-as-product false positives.
3. **Negative Event Filtering:** Heuristic filters reject financial disclosures (e.g., quarterly earnings releases, senior note offerings, sales guidance) and executive governance events (e.g., CFO appointments, board retirements). Generic nouns (`solutions`, `new products`) are blocked.
4. **Length Verification:** Product descriptions are strictly validated against a 179-character ceiling (`0 < len(desc) < 180`), adhering to the exact assignment constraint.

6. Empirical Execution Telemetry & System Performance

The complete automated pipeline processed all 100 selected SEC filings in a total execution session duration of 7.37 minutes across 135 structured LLM requests [5].

Pipeline Performance Metric	Execution Value	System Target
Total Candidate Companies Ingested	**20**	20 entities
Filings Discovered & Successfully Completed	**100 / 100 (100%)**	100 unique filings
Total Structured LLM Model Calls	**135**	Dynamic (≤12 / filing)
Candidate Product Extractions Flagged for Review	**15**	Audit queue
Provisional Validated Product Rows Generated	**7**	Provisional export

Pipeline Performance Metric	Execution Value	System Target
Active Model GPU VRAM Footprint	**3.61 GiB**	T4 / A100 runtime
Total Pipeline Execution Duration	**7.37 minutes**	4.5 hour budget

> **Core Ingestion Finding:** 100% of discovered filings were successfully parsed within documented exhibit scope. The model achieved zero fatal crashes or JSON validation failures, proving the efficacy of strict JSON schema constraints coupled with GPU-accelerated local Ollama inference.

7. Provisional Extracted Product Portfolio

Below are the seven provisional commercial releases successfully passing all automated grounding and schema validation checks:

Company / Ticker	Filing Timestamp (UTC)	Extracted Product Name	Extracted Description (<180 chars)
Azenta, Inc. (AZTA)	2025-12-29 14:25:46	B Medical Systems business	Azenta announced the sale of its B Medical Systems business to THELEMA S.À R.L. for US$63 million, with closing expected by March 31, 2026.
Cerence Inc. (CRNC)	2025-11-19 21:06:45	xUI platform	Cerence xUI platform, with first xUI-powered cars expected to hit roads in 2026.
IPG PHOTONICS (IPGP)	2025-08-05 12:00:47	directed energy system for counter-UAV applications	A directed energy system for counter-UAV applications, launching to unlock significant value for customers.
DIODES INC (DIOD)	2025-08-11 21:00:28	PCI Express 6.0 ReDrivers, clock buffers, and clock muxes	Introduced new PCI Express 6.0 ReDrivers, clock buffers, and clock muxes seeing strong momentum in server and data center applications.
DIODES INC (DIOD)	2025-08-11 21:00:28	MIPI D-PHY ReDrivers	Rapid adoption of MIPI D-PHY ReDrivers in robot, drones, mixed reality, and embedded MMC switches in gaming console applications.
Navitas Semi (NVTS)	2025-11-03 21:11:58	100V and 650V discrete GaNFast™ FETs	A newly introduced portfolio of 100V and 650V discrete GaNFast™ FETs enabling next-generation 800V DC AI factory power architecture.
Navitas Semi (NVTS)	2025-11-03 21:11:58	2.3kV and 3.3kV high-voltage SiC modules	Sampling new 2.3kV and 3.3kV high-voltage SiC modules to leading energy-storage and grid-infrastructure customers.

8. Qualitative Error Analysis & Methodological Limitations

Auditing the pipeline's provisional outputs and review queues yields crucial insights into corporate NLP extraction challenges:

- **Semantic False Positives:** The extraction of *"B Medical Systems business"* (Azenta / AZTA) represents a critical semantic false positive. While verbatim source grounding and release vocabulary were present, the disclosure actually announced a corporate divestiture ($63 million sale of a business unit), not an organic product release. Lexical overlap cannot substitute for deep semantic legal reasoning.
- **Boundary & Exhibit Truncation:** Bounding scope to the primary 8-K and Exhibit 99.1 prevents compute bottlenecks but risks omitting releases published in technical annexes (Exhibit 99.2) or Item 1.01 commercial contract schedules.
- **Zero-Row Legitimacy:** Out of 100 examined filings, 93 yielded zero valid product rows. In regulatory accounting, the vast majority of Form 8-K submissions concern governance (Item 5.02), financing (Item 3.02), or periodic earnings summaries (Item 2.02). A sparse output is structurally expected; injecting synthetic entries would compromise regulatory integrity.

- **Human-in-the-Loop Audit Separation:** In compliance with reproducible evaluation standards, unverified machine outputs (`exports/products.csv`) remain segregated from formally approved human audits (`reviewed/products.csv`). Because no manual approval override was committed during the latest automated run, the reviewed export remains intentionally unpopulated.

9. Reproducibility & Checkpointing Protocols

The architecture guarantees complete state persistence across execution environments. All runtime configurations, raw inference JSON payloads, intermediate text batches, and filing records are hashed (`sha256`) and written atomically to persistent Google Drive storage. Interrupted sessions automatically resume from the latest uncommitted filing batch without re-querying completed items. The full code repository, configuration schemas, and data artifacts are publicly archived at the repository linked above.

10. Formal References

1. Securities and Exchange Commission (SEC), "EDGAR Application Programming Interfaces & Access Guidelines," *SEC.gov Developer Resources*, 2025. Available: https://www.sec.gov/edgar/sec-api-documentation
2. L. Richardson, "Beautiful Soup: Tool for parsing HTML and XML documents," *Crummy Software*, 2024. Available: https://www.crummy.com/software/BeautifulSoup/bs4/doc/
3. Ollama Project, "Structured JSON Schema Outputs & Local Model Serving Documentation," *Ollama AI*, 2025. Available: https://docs.ollama.com/capabilities/structured-outputs
4. Google Research, "Google Colaboratory Resource Allocations and Hardware Specifications FAQ," *Google Research Documentation*, 2025. Available: https://research.google.com/colaboratory/faq.html
5. C. Chachengwa, "End-to-end Python pipeline extracting new product announcements from SEC 8-K filings via EDGAR API and Ollama LLMs into structured CSV data," *LLM-Document-Analysis Suite*, 2026. Accessible via project archive: GitHub / LLM-Document-Analysis